

Device Connection	Cable Type	Connector	Why It Is Suitable
Gaming Console → TV/Monitor	HDMI 2.1	HDMI Type-A	Supports 4K at 120 Hz , 8K, HDR, and high-quality audio with very low latency. Ideal for PlayStation 5 and Xbox Series X.
Gaming PC → Gaming Monitor	DisplayPort 1.4/2.1	DisplayPort	Provides higher refresh rates (144 Hz, 240 Hz, or more) and excellent gaming performance.
Router/Switch → Consoles/PCs	Cat6 Ethernet Cable	RJ45	Offers 1 Gbps to 10 Gbps wired networking with lower latency than Wi-Fi, ensuring smooth multiplayer gaming.
Laptop → Display (if supported)	USB Type-C (DisplayPort Alt Mode)	USB-C	Carries video, audio, and data through a single cable, reducing cable clutter.

Why These Choices?

1. HDMI 2.1

- Best for connecting gaming consoles to large TVs.
- Supports:
 - 4K @ 120 Hz
 - 8K @ 60 Hz
 - HDR (High Dynamic Range)
 - Variable Refresh Rate (VRR)
 - Enhanced Audio Return Channel (eARC)
- Ensures smooth gameplay and excellent picture quality during the IPL event.

2. DisplayPort

- Preferred for gaming PCs.
- Supports higher refresh rates than HDMI in many monitor setups.
- Ideal for competitive gaming because it minimizes screen tearing and lag.

3. Cat6 Ethernet with RJ45

- Provides a stable, wired internet connection.
- Reduces latency compared to Wi-Fi.
- Essential for online multiplayer games and live streaming without interruptions.

4. USB Type-C

- Useful for modern laptops or portable gaming devices.
- One cable can carry video, audio, data, and sometimes power.